

Complexity vs time-dependence: model reduction for classical and quantum mechanics

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In any branch of science, the ability to simulate the evolution of large dynamical models is of paramount importance but, whenever the model of interest is too large to be simulated exactly, one has to recur to model reduction methods and other approximation techniques to be able to simulate an approximate evolution of the process. In this work we explore a different approach to what is typically done to reduce the size of large models in control engineering: instead of looking for a linear time-invariant model we look for a linear time-varying model whose dimension is the same of the process we want to reproduce. Such a problem is typical in quantum mechanics for the derivation of time-convolutionless master equations.

Here we propose a novel method to compute an approximation of the linear time-varying generator that provides an alternative to the typical derivation of quantum master equations. The proposed approximation is polynomial in the time variable and the derivation is based on the Taylor expansion of the exponential map and a novel result on the computation of the time-ordered exponential for time-dependent generators that are polynomial in time. The resulting generator approximate the exact evolution for small times, does not require a weak coupling assumption and is guaranteed to provide a CPTP semigroup at the lowest orders. Importantly, the coefficients of the polynomial can be computed through a recursive formula which numerically gives easy access to higher order approximations. The proposed method is tested against prototypical models of including quantum models of interest: a dephasing spin-boson model a central spin model and an Ising spin chain.

I. INTRODUCTION

Whenever a dynamical model is too large to be simulated efficiently, one should recur to model reduction techniques to find a smaller model that is simulatable and that retains fundamental characteristics of the original one. Typical methods considered in control systems theory to perform model reduction such as balanced truncation [1, 2] and moment matching [3, 4] look for linear time invariant (LTI) models that approximate the evolution of the original model. Here, we consider a different approach: we look for a linear time-varying (LTV) model of a fixed size that approximate the evolution.

Such an approach is motivated by physics. To obtain realistic models of quantum systems, one shall consider the effects that the interaction between the system and its environment have on the evolution of the system. Such an interaction generates important behaviors such as dissipation and dephasing hence modelling these effects precisely is a mandatory task to study open quantum systems [5]. Unfortunately, modelling the entire system is only doable in theory as the scaling of the Hilbert space prohibits any simulation or analysis of the joint system–environment even for small sizes of the environment. This fact implies that in order to study the effect of the coupling with the environment, one needs to reduce the model and aim for an approximate description of the system that hopefully captures the interesting features of the original model.

A common method to reduce the joint system–environment model is known in the physics literature with the name of time-convolutionless master equations (TCL ME) [6, 7]. Such a method aims at obtaining a linear time-varying model (LTV) that replicates the effects of the environment by leveraging an expansion in

the strenght of the coupling between the system and environment, which is typically assumed to be small. For more details see e.g. [5] and references therein.

In this work we propose an alternative method to derive a similar approximation. In particular, instead of performing an expansion in one of the system’s parameters, we perform an expansion in the time variable t . This allows us to derive a reduced model that is time-varying but only polynomial in t . Under these assumptions we are able to propose a recursive formula for the coefficients of the polynomial generator that offers an approximation whose error scales as $O(t^N)$ where N is the order of the polynomial.

Note that, while time-dependent dynamical systems are typically more difficult to study, such an approach finds an important application in the simulation of very large systems as the Runge-Kutta methods allow to simulate time-dependent systems [8].

When we apply the methods proposed in this work to quantum systems, we are able to prove that, in the common case of purely Hamiltonian evolution, the reduced polynomial generator at first and second order is in Lindblad form for all times t .

It is important to point out that the method proposed in this work is not intended to substitute the perturbative approach at deriving the TCL ME [6, 7] but rather to provide an alternative that can be of use whenever the main assumptions of the standard method do not hold, e.g. in the strong coupling regime or whenever the time-ordered integrals are difficult to assess.

For the reader’s convenience we next summarize the main contributions of this work:

- In Sec. III we propose the polynomial approximation of the evolution of LTI systems for general linear autonomous (not necessarily quantum)

models and present the recursive equation that determines the coefficients of the polynomial. We also test the proposed solution on a linear model and compare its results against the truncation of the exponential series.

- In Sec. IV we connect the proposed method with the derivation of TCL ME and, importantly, we discuss the connections between the recursive formula we obtain and other similar recursive equations known in the physics literature [9–12] by highlighting how the expansion in t greatly simplifies the associated calculations. Furthermore, we leverage a few recent results [13, 14] to prove that, when the original dynamics is purely Hamiltonian the proposed approximation at first and second order provides a CPTP (two-parameter) semi-group.
- In Sec. V we test the proposed procedure on several prototypical quantum examples: a dephasing spin boson model, a central spin model and an Ising spin chain.

Throughout the paper we include pedagogical examples that intuitively explain the most involved proofs and highlight both the strengths and weaknesses of the proposed approximation procedure, namely: the recursive formula used to compute the terms of the generators at different orders makes the procedure numerically easier to access at higher orders than the alternatives; allows for an initial model that is in Lindblad form (not only purely Hamiltonian); does not require the model to be put in rotating frame; and does not rely on a weak coupling assumption hence it allows to study systems even in strong coupling regimes. On the other hand, the proposed reduction procedure relies on an expansion in time and thus provides a good approximation only for small times. While this aspect is limiting, similar restriction also hold in the well known approximations of the time-convolutionless master equation proposed by [6].

II. PROBLEM SETTING AND CONNECTION WITH THE LITERATURE

A. Problem setting

In this work we assume to be given a LTI stable autonomous model of the type

$$\dot{x}_t = Lx_t \quad (1)$$

where $x_t \in \mathbb{C}^n$ is the system's state and $L \in \mathbb{C}^{n \times n}$ is the state matrix or generator. Such a generator defines a one-parameter (Markovian) semigroup, i.e. $\{\Lambda_t \equiv e^{Lt}\}_{t \geq 0}$ such that $\Lambda_{t+s} = \Lambda_t \Lambda_s$, for all $t, s \geq 0$.

We next assume that, for some reason dictated by the application at hand, we are actually not interested in reproducing all the degrees of freedom contained in the system's state x_t , but we are only interested in the degrees

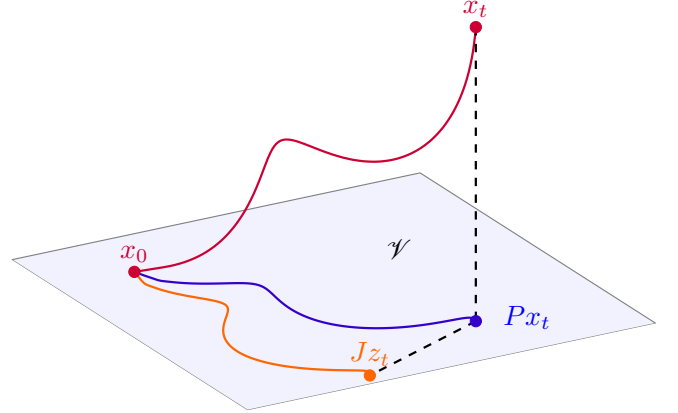


Figure 1: Graphical representation of the approximation problem. The pale blue plane represents the subspace $\mathcal{V}$ onto which P projects. The red curve represents the trajectory of the original model x_t , which starts in $\mathcal{V}$ at $t = 0$. The blue curve represents the projection of the true trajectory Px_t , i.e. the one we want to approximate. Finally, the orange curve represents the approximate evolution we aim to find, i.e. Jz_t .

of freedom contained on a given subspace $\mathcal{V} \subseteq \mathbb{C}^n$. Let then $P \in \mathbb{C}^{n \times n}$, be a projector $P^2 = P$ onto $\mathcal{V} = \text{Im}P$. Note that P does not need to be orthogonal (with respect to the standard inner product $\langle \cdot, \cdot \rangle$), i.e. $P^\dagger \neq P$. For future convenience, let us define $Q \equiv \mathbb{1} - P$ where $\mathbb{1}$ denotes the identity matrix, let us denote by m the dimension of $\mathcal{V}$ and let us factorize P into two linear maps: the reduction $R \in \mathbb{C}^{m \times n}$ and the injection $J \in \mathbb{C}^{n \times m}$ such that $P = JR$ and $RJ = \mathbb{1}_m$ the identity in $\mathbb{C}^{m \times m}$.

We further make a simplifying assumption that is quite typical in these settings, see e.g. [5]: We assume that $x_0 \in \mathcal{V}$, or equivalently $Px_0 = x_0$.

Summarizing, in this work we are interested in deriving a reduced model that is capable of reproducing the trajectories

$$Pe^{Lt}x_0, \quad \forall x_0 \in \mathcal{V}, \quad \forall t \geq 0 \quad (2)$$

or, equivalently,

$$Re^{Lt}J\tilde{x}_0, \quad \forall \tilde{x}_0 \in \mathbb{C}^m, \quad \forall t \geq 0. \quad (3)$$

Here however, we take a different route to the standard approach to model reduction used in the control community, e.g. balanced truncation [1, 2] and moment matching [3, 4]. In particular, instead of aiming to reduce the trajectories of interest by looking for an LTI model of dimension larger than m , we look for an LTV model of dimension m that approximate the trajectories of interest. Note that while this approach is not common in the

control literature this has been studied in the physics literature for the derivation of TCL ME as detailed in Sec. IV A.

Let us then define a state $z_t \in \mathbb{C}^m$ whose evolution is given by an LTV model of the type

$$\begin{cases} \dot{z}_t = F_t z_t \\ z_0 = R x_0 \end{cases} \quad (4)$$

where F_t is a time-dependent generator. The evolution of such an LTV model is given by the two-parameter (non-Markovian) semigroup $\{\Upsilon_t \equiv \mathbb{T}e^{\int_0^t F_s ds}\}_{t \geq 0}$, where $\mathbb{T}$ denotes the time-ordering operator, for which the composition rule $\Upsilon_{t,0} = \Upsilon_{t,s}\Upsilon_{s,0}$ for all $0 \leq s \leq t$ holds.

Our aim is then to construct F_t so that $x_t \approx R x_t$. In other words we want to determine F_t so that

$$\mathbb{T}e^{\int_0^t F_s ds} \approx R e^{L t} J.$$

The sense in which the approximation $\approx$ is intended shall be defined more precisely later on in the paper. A graphical representation of the approximation problem is given in Fig. 1.

B. Time-convolutionless master equation

Before moving to the method proposed in this work for computing the time dependent generator F_t we shall discuss the derivation of time-convolutionless master equations, which is the standard method found in the physics literature [6, 7, 15, 16] to solve the issue at hand. The derivation is particularly elegant and makes use of the Lagrange equation. We report it here for the interested reader.

Dividing the evolution of x_t in the evolution of $P x_t$ and that of $Q x_t$ we have

$$\frac{d}{dt} P x_t = P L P x_t + P L Q x_t, \quad (5)$$

$$\frac{d}{dt} Q x_t = Q L P x_t + Q L Q x_t. \quad (6)$$

Solving the second equation for $Q x_t$ (using the Lagrange equation) we obtain

$$Q x_t = e^{Q L Q t} Q x_0 + \int_0^t e^{Q L Q s} Q L P x_{t-s} ds.$$

By substituting this equation into Eq. (5) we obtain the famous Nakajima-Zwanzig equation [17, 18]:

$$\begin{aligned} \frac{d}{dt} P x_t &= P L P x_t + P L e^{Q L Q t} Q x_0 \\ &\quad + \int_0^t P L Q e^{Q L Q s} Q L P x_{t-s} ds. \end{aligned} \quad (7)$$

We then recall that $x_{t-s} = e^{-L s} x_t$ so that

$$Q x_t = e^{Q L Q t} Q x_0 + \int_0^t e^{Q L Q s} Q L P e^{-L s} ds \cdot (P + Q) x_t.$$

Now, defining

$$M_t \equiv \mathbb{1} - \int_0^t e^{Q L Q s} Q L P e^{-L s} ds$$

we have

$$M_t Q x_t = e^{Q L Q t} Q x_0 + (\mathbb{1} - M_t) P x_t.$$

If we then assume that M_t is invertible we obtain

$$Q x_t = M_t^{-1} e^{Q L Q t} Q x_0 + (M_t^{-1} - \mathbb{1}) P x_t.$$

Substituting this in Eq. (5) we obtain

$$\begin{aligned} \frac{d}{dt} P x_t &= P L P x_t + P L M_t^{-1} e^{Q L Q t} Q x_0 \\ &\quad + P L (M_t^{-1} - \mathbb{1}) P x_t \\ &= P L M_t^{-1} e^{Q L Q t} Q x_0 + P L M_t^{-1} P x_t. \end{aligned} \quad (8)$$

Note that under the assumptions that $P x_0 = x_0$ we have that $Q x_0 = 0$ and thus Eq. (8) gets simplified to

$$\frac{d}{dt} P x_t = \underbrace{P L M_t^{-1} P}_{\equiv K_t} \rho_t \quad (9)$$

which, separating the Projector P into its factors $P = J R$, gets simplified to

$$\frac{d}{dt} \tilde{x}_t = \underbrace{R L M_t^{-1} J}_{\equiv F_t} \tilde{x}_t. \quad (10)$$

A few comments on this solution are in order: i) The invertibility of M_t needs to be verified and typically only holds for small t , see e.g. [5]. ii) Note that, in order to compute F_t (or equivalently K_t) one needs to first compute the convolution integral that defines M_t and then to compute the inverse of M_t . Both these tasks are challenging (numerically and analytically) and it is generally considered that solving this integral and inverse is as difficult as solving the original model (1), see e.g. [19]. Nevertheless, Eq. (8) provides an excellent starting point to compute an approximation of the reduced evolution since, to circumvent these numerical issues, K_t is typically approximated by expanding it as an expansion in the coupling strength ε and by computing the terms of the expansion as ordered cumulants [5, 9, 20]. In many cases of interest this approximation of the TCL-ME provides a good approximation of the open quantum system dynamics, both in the long-time and short-time regime. It is however important to note that there is no guarantee that such an approximation holds also for long times. As an example of this, see [19] where it is shown that TCL2 and TCL4 (approximations at orders $N = 2, 4$) do not approximate the true dynamics in the long time regime. We refer the reader to Sec. IV B for further connections with the physics literature.

III. POLYNOMIAL APPROXIMATIONS OF THE TIME-DEPENDENT GENERATOR

This work leverages a key assumption: we assume that the time dependent generator F_t is polynomial in t , i.e. $F_{t,N} \equiv \sum_{k=0}^N t^k F_{(k+1)}$. This assumption is clearly restrictive, but it greatly simplifies the computation of the time-ordered exponential $\mathbb{T}e^{\int_0^t F_{s,N} ds}$ as shown in the next key result.

Theorem 1. *Let us assume that $F_{t,N} = \sum_{k=0}^N t^k F_{(k+1)}$. Then the time-ordered exponential can be written as*

$$\mathbb{T}e^{\int_0^t F_{s,N} ds} = \sum_{k=0}^{+\infty} t^k E_{(k)} \quad (11)$$

where $E_{(0)} = \mathcal{I}$ and

$$E_{(k)} = \frac{1}{k} \sum_{s=1}^k F_{(s)} E_{(k-s)} \quad (12)$$

and where we used the convention $F_{(k)} = 0$ for all $k > N$.

Proof of this result and an explanatory example are given in Appendix A.

Note that Eq. (12) provides a closed formula for the coefficients $E_{(k)}$ of the expansion of $\mathbb{T}e^{\int_0^t F_{s,N} ds}$ as a function of the coefficients $F_{(k)}$. Furthermore, only the terms $\{E_{(\ell)}\}_{\ell=1}^{k-1}$ appear in Eq. (12), which means that this equation can be used iteratively to compute each $E_{(k)}$.

Theorem 1 provides the main intuition on how to construct the polynomial time dependent generator $F_{t,N}$: Eq. (11) provides an expansion for $\mathbb{T}e^{\int_0^t F_{s,N} ds}$ in the time variable t . We can then expand e^{tL} in t and construct $F_{(k)}$ so that the terms of $\mathbb{T}e^{\int_0^t F_{s,N} ds}$ match the terms of $Re^{tL}J$ by orders of t . This approach is similarly in principle to adiabatic elimination techniques [21] but where instead of matching orders of ε we match orders of t . The resulting procedure, which represents the main contribution of this work, is stated next.

Theorem 2. *Let us define*

$$F_{t,N} \equiv \sum_{k=0}^N t^k F_{(k+1)} \quad (13)$$

where

$$F_{(k)} \equiv k \frac{RL^k J}{k!} - \sum_{h=1}^{k-1} F_{(k-h)} \frac{RL^h J}{h!}. \quad (14)$$

Then:

$$Re^{tL}J - \mathbb{T}e^{\int_0^t F_{s,N} ds} = O(t^{N+1}). \quad (15)$$

Proof. First of all we can use Theorem 1 to write

$$\mathbb{T}e^{\int_0^t F_{s,N} ds} = \sum_{k=0}^{+\infty} t^k E_{(k)}$$

where $E_{(k)} = \frac{1}{k} \sum_{s=1}^k F_{(s)} E_{(k-s)} = \frac{1}{k} \sum_{h=0}^{k-1} F_{(k-h)} E_{(h)}$ and where we reversed the order of the summation by imposing $h = k - s$. We next prove that, for $1 \leq k \leq N$, we have $E_{(k)} = \frac{RL^k J}{k!}$ by induction. The case $k = 1$ is trivial as $E_{(1)} = F_{(1)} E_{(0)} = RLJ$. Let then assume that $E_{(h)} = \frac{RL^h J}{h!}$ for all $h < k$ and let us prove that $E_{(k)} = \frac{RL^k J}{k!}$. Then

$$\begin{aligned} E_{(k)} &= \frac{1}{k} \sum_{h=0}^{k-1} F_{(k-h)} E_{(h)} = \frac{F_{(k)}}{k} + \frac{1}{k} \sum_{h=1}^{k-1} F_{(k-h)} E_{(h)} \\ &= \frac{F_{(k)}}{k} + \frac{1}{k} \sum_{h=1}^{k-1} F_{(k-h)} \frac{RL^h J}{h!} = \frac{RL^k J}{k!} \end{aligned}$$

where we simply substituted the definition of $F_{(k)}$ given in eq. (14).

By leveraging the definition of the exponential map e^{Lt} we can expand the term $Re^{Lt}J$ to obtain

$$Re^{Lt}J = \sum_{k=0}^{+\infty} \frac{t^k}{k!} RL^k J.$$

Combining the two terms we obtain

$$\begin{aligned} Re^{Lt}J - \mathbb{T}e^{\int_0^t F_{s,N} ds} &= \sum_{k=0}^{+\infty} t^k \left(\frac{RL^k J}{k!} - E_{(k)} \right) \\ &= \sum_{k=0}^N t^k \left(\cancel{\frac{RL^k J}{k!}} - E_{(k)} \right) \\ &\quad + O(t^{N+1}) \end{aligned}$$

which concludes the proof. $\square$

We shall observe a few key aspects. First, Eq. (15) finally gives sense to the approximation symbol $\approx$ used in Sec. II i.e. an approximation of the type $O(t^{N+1})$. This implies that the approximation holds only for small times. Second, it is important to notice that the right-hand side of equation (14) depends on $RL^k J$ and $\{F_{(s)}\}_{s=1}^{k-1}$. This means that equation (14) can be used iteratively to construct the terms $F_{(k)}$ starting from $k = 1$ and growing up to $k = N$. For the reader's convenience the first few terms of the series are reported next:

$$\begin{aligned} F_{(1)} &= RLJ \\ F_{(2)} &= RL^2 J - F_{(1)} RLJ = RL^2 J - (RLJ)^2 \\ F_{(3)} &= \frac{RL^3 J}{2} - F_{(2)} RLJ - F_{(1)} \frac{RL^2 J}{2} \\ &= \frac{RL^3 J}{2} - \frac{F_{(1)} F_{(2)}}{2} - \frac{F_{(1)}^3}{2} - F_{(2)} F_{(1)} \end{aligned}$$

It is interesting to notice a couple of points about the first few terms: i) As one would expect, the term of order one $F_{(1)}$ coincides with the reduction of the original dynamics which is often used as rough time-independent approximation of the reduced evolution [1]; ii) The second order

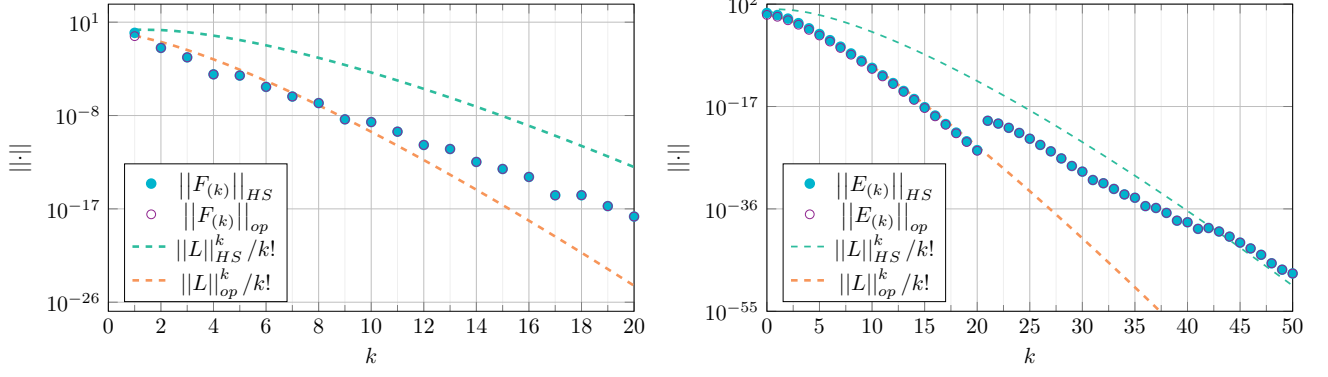


Figure 2: Operator and Hilbert-Schmidt norm of the first 20 $F_{(k)}$ terms (left) and of the first 50 $E_{(k)}$ terms when $N = 20$ (right).

term does not only contain the reduction of the generator squared RL^2J but also contains a correction term depends on the first order term $F_{(1)}$.

Lastly, it is important to note that similar (yet distinct) recursive formulas to derive similar approximations have been found in the physics literature [9–12]. We refer the reader to Sec. IV B where we discuss connections and differences between these works.

A. Linear testbed

To showcase the capability of the proposed method, we here consider a simple numerical example. Specifically, we consider a random autonomous and stable system of size $n = 20$. Let then A be a random uniform matrix in $\mathbb{C}^{n \times n}$, i.e. $[A]_{i,j} \sim \mathcal{U}([0, 1])$ and let $B = A - \lambda_{\text{Max}} \mathbb{1}_n$ where λ_{Max} is the absolute value of largest eigenvalue of A . The dynamic's generator $L \in \mathbb{C}^{n \times n}$ is then obtained as $L = \frac{1}{2} \frac{B}{\|B\|_{\text{op}}}$ where $\|\cdot\|_{\text{op}}$ denotes the operator norm. This ensures that L is simply stable and has operator norm $\|L\|_{\text{op}} = \frac{1}{2}$.

The projector P is then constructed through $R = [\mathbb{1}_{m \times m} \mid 0_{m \times n-m}]$ and $J = R^\dagger$ to obtain

$$P = \begin{bmatrix} \mathbb{1}_{m \times m} & 0_{m \times n-m} \\ 0_{m \times n-m} & 0_{n-m \times n-m} \end{bmatrix}.$$

By using Eq. (14) we can then compute the terms $F_{(k)}$ iteratively for $k = 1, \dots, N$. Fig. 2 (left) shows both the Hilbert-Schmidt and operator norm of the terms $F_{(k)}$ for $k = 1, 20$. Interestingly, the Hilbert-Schmidt norm, $\|F_{(k)}\|_{\text{HS}}$, is bounded by $\frac{\|L\|_{\text{HS}}^k}{k!}$, indicating that the series $F_{t,N}$ should be converging for $N \rightarrow \infty$. Curiously, the same does not hold for the operator norm. Fig. 2 (right) instead shows the norm of the terms $E_{(k)}$ where we fixed $N = 20$ and $F_{(k)} = 0 \forall k > 20$. Interestingly for $k \leq N$ $\|E_{(k)}\|$ is bounded by $\frac{\|L\|_{\text{op}}^k}{k!}$ while for $k > N$ this is no longer the case.

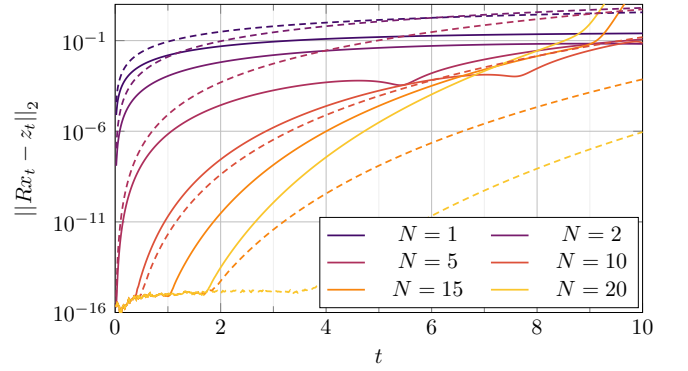


Figure 3: The continuous lines represent the error norm between the exact reduced evolution Rx_t and the approximated reduced evolution z_t for different values of the approximation order N . The dashed curves represent an approximation obtained by truncating the Taylor expansion of the exponential for different values of the truncation order N .

Fig. 3 shows the error norm $\|Rx_t - z_t\|_2$ versus time. In this simulation Rx_t was simulated by reducing the original model, i.e. $Rx_t = Re^{Lt}x_0$ while z_t is obtained by simulating the time-dependent linear differential equation $\dot{z}_t = F_{t,N}z_t$, for different values of N , by using an approximation of the equation given in Theorem 1, i.e. $\mathbb{T}e^{\int_0^t F_{s,N} ds} \approx \sum_{k=0}^{100} t^k E_{(k)}$. One can note that increasing the degree of approximation N decreases the error committed by the approximated model.

A natural alternative numerical approximation that comes to mind is given by truncating the exponential expansion at a given order N , i.e. $Rx_t \approx \sum_{k=0}^N \frac{t^k}{k!} RL^k J Rx_0$. Note that such an approximation has numerical complexity almost identical to that of the approximation proposed in this work as in both cases one need to compute L^N . Furthermore, from a theoretical standpoint, both cases commit an error of the order of $O(t^{N+1})$. To test this fact numerically, in Fig. 3 we included the actual error committed by these two ap-

proximations at the same orders of approximation: the continuous line denotes the error committed by the procedure proposed here while the dashed line is the error committed by the truncation of the exponential. One can observe that for lower orders ($N = 1, 2, 5$) the procedure proposed here yields better results, while at higher orders this is not the case.

IV. APPLICATION TO QUANTUM MECHANICS AND THE COMPLETE POSITIVITY OF THE FIRST AND SECOND ORDER APPROXIMATIONS

In this section we apply the results derived in the Sec. III to quantum mechanical models, the application that originally motivated this work.

A. The problem in quantum mechanics

Let us consider a quantum system defined on a Hilbert space $\mathcal{H}$, and let $\mathfrak{B}(\mathcal{H})$ denote the set of bounded operators acting on $\mathcal{H}$. Unless differently specified, we only consider finite-dimensional systems, i.e. $\dim(\mathcal{H}) < \infty$, $\mathcal{H} \simeq \mathbb{C}^n$ and $\mathfrak{B}(\mathcal{H}) \simeq \mathbb{C}^{n \times n}$. The state of such a system is described by a density operator $\rho \in \mathfrak{D}(\mathcal{H})$ with $\mathfrak{D}(\mathcal{H}) = \{\rho \in \mathfrak{B}(\mathcal{H}) | \rho = \rho^\dagger \geq 0, \text{tr} \rho = 1\}$. Here, we assume that the state of the system evolves according to the Von-Neumann master equation

$$\dot{\rho}_t = \mathcal{L}(\rho_t) \quad (16)$$

where $\rho_0 \in \mathfrak{D}(\mathcal{H})$ is the system's initial condition, and $\mathcal{L}$ is the celebrated Lindblad (or GKLS) generator [22, 23]:

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_k \mathcal{D}_{L_k}(\rho) \quad (17)$$

where $H = H^\dagger$ is the system's Hamiltonian and $\mathcal{D}_L(\rho) = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$ represents the system's dissipation toward a Markovian environment represented by a set of operators $\{L_k\}$ called noise operators. The evolution of the quantum system is then described by the exponential map $\rho_t = e^{\mathcal{L}t}(\rho_0)$. Importantly, $e^{\mathcal{L}t}$ generates a quantum dynamical semigroup, i.e. $\{\Lambda_t \equiv e^{\mathcal{L}t}\}_{t \geq 0}$ a one-parameter semigroup of completely positive (CP) and trace preserving maps (TP) also known as Markovian dynamics. In the following we will say that a superoperator is of *Lindblad-type* if it can be put in the form of Eq. (17).

Note that typically, in the derivation of TCL ME [6, 7], the initial “big” model is considered to be Hamiltonian, i.e. $\mathcal{L}(\cdot) = -i[H, \cdot]$. However, mathematically, there is no reason to restrict to this case hence, in this work, we start with the general assumption that $\mathcal{L}$ is a generator of Lindblad form and then focus on the Hamiltonian case when necessary.

In many situation of interest in quantum physics, relevant projectors are represented by CPTP ones. We thus assume to be given a CPTP, not necessarily orthogonal (w.r.t. $\langle \cdot, \cdot \rangle_{HS}$), projector $\mathcal{P} : \mathfrak{B}(\mathcal{H}) \rightarrow \mathfrak{B}(\mathcal{H})$, $\mathcal{P}^2 = \mathcal{P}$, whose image will be denoted by $\mathcal{V} \equiv \text{Im} \mathcal{P}$. This provides a generalization to the superoperator projector setting considered by [19, 20]. Furthermore, for the sake of simplicity, we shall assume that $\mathcal{P}$ admits a full rank fixed point, i.e. $\exists \rho \in \mathfrak{D}(\mathcal{H})$, $\rho > 0$ such that $\mathcal{P}(\rho) = \rho$. As we shall see in more details later (in Sec. IV C) such a projector is in fact a conditional expectation in quantum probability theory [24, 25] and enjoys some useful and well known properties [13, 26–31]. For the moment however, it is sufficient to know that $\mathcal{P}$ can be decomposed in two factors $\mathcal{P} = \mathcal{J}\mathcal{R}$ which take the name of reduction and injection maps $\mathcal{R} : \mathfrak{B}(\mathcal{H}) \rightarrow \mathfrak{B}(\mathcal{K})$ and $\mathcal{J} : \mathfrak{B}(\mathcal{K}) \rightarrow \mathfrak{B}(\mathcal{H})$ such that $\mathcal{R}\mathcal{J} = \mathcal{I}_{\mathcal{K}}$ the identity superoperator, $\text{Im} \mathcal{J} = \mathcal{V}$, $\text{Im} \mathcal{R} = \mathcal{A}$ and where $\mathcal{A} \subseteq \mathfrak{B}(\mathcal{K})$ is a $*$ -subalgebra of $\mathfrak{B}(\mathcal{K})$ for some Hilbert space $\mathcal{K}$, $\dim(\mathcal{K}) \leq \dim(\mathcal{H})$.

To provide some intuition on these mathematical objects we anticipate the reduction and injection maps we are going to use in the applications of Sec. V:

- Assume a bipartite case, i.e. $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$. We can then consider $\mathcal{P}(\rho) = \text{tr}_B(\rho) \otimes \tau$ for some $\tau \in \mathfrak{D}(\mathcal{H}_B)$, $\tau > 0$ to be our CPTP projector. Then, by choosing $\mathcal{K} = \mathcal{H}_S$ and $\mathcal{A} = \mathfrak{B}(\mathcal{H}_S)$ we have the maps $\mathcal{R}(\rho) = \text{tr}_B(\rho)$ and $\mathcal{J}(\mu) = \mu \otimes \tau$.
- Another CPTP projector we might want to consider is the projector onto the diagonal matrix $\mathcal{P}(\rho) = \sum_{j=0}^{n-1} |j\rangle\langle j| \rho |j\rangle\langle j|$ in which case $\mathcal{K} = \mathcal{H}$, $\mathcal{A}$ is the commutative algebra given by $\mathcal{A} \equiv \text{span}\{|j\rangle\langle j|, j = 0, \dots, n-1\} \simeq \mathbb{C}^n$, and $\mathcal{R}(\rho) = \sum_{j=0}^{n-1} |j\rangle\langle j| \rho |j\rangle\langle j|$ in which case $\mu = \mathcal{R}(\rho) \in \mathbb{C}^n$, and $\mathcal{J}(\mu) = \sum_{j=0}^{n-1} |j\rangle\langle j| \mu |j\rangle\langle j|$ factorize $\mathcal{P}$.

The reduction map $\mathcal{R}$ determines the reduced state – the one we want to keep track of: Let $\eta_t \in \mathfrak{D}(\mathcal{K})$ be defined as $\eta_t = \mathcal{R}(\rho_t)$ for all $t \geq 0$. Under the assumption $\mathcal{P}(\rho_0) = \rho_0$, the evolution of the reduced state is thus given by

$$\eta_t = \mathcal{R}e^{\mathcal{L}t}(\rho_0) = \mathcal{R}e^{\mathcal{L}t}\mathcal{J}(\eta_0).$$

Theorem 2 then offers a method to approximate this evolution with a time-dependent generator $\mathcal{F}_{t,N}$ that is polynomial in time t , i.e. $\mathcal{F}_{t,N} = \sum_{k=0}^N t^k \mathcal{F}_{(k+1)}$ and such that

$$\left\| \mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds} - \mathcal{R}e^{\mathcal{L}t}\mathcal{J} \right\| = O(t^{N+1}).$$

Desirable properties for $\mathcal{F}_{t,N}$ certainly include the fact that we would like the semigroup $\{\Upsilon_t \equiv \mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds}\}_{t \geq 0}$ to be a set of CPTP maps for all $t \geq 0$. Note that necessary conditions to impose on $\mathcal{F}_t$ so that $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds}$ is CPTP for all $t \geq 0$ are not known [32]. Nevertheless in the next few sections we shall see that under mild assumptions the first two orders of the approximation ensure that this desirable property is preserved.

B. Connection with the physics literature

In the typical setting of TCL ME derivation, the system is considered to be bipartite, i.e. $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$ and the dynamics purely Hamiltonian, i.e. $\mathcal{L}(\rho) = -i[H, \rho]$ with Hamiltonian $H = H_0 + \varepsilon V$ where H_0 is the free Hamiltonian of the system and bath, V is the interaction Hamiltonian and ε is the coupling strength. Under these assumptions, and assuming that the coupling strength is small, the time dependent generator $\mathcal{K}_t$ is computed by performing a perturbative expansion in the coupling parameter. The time-dependent generator is then expressed as

$$\mathcal{K}_t = \sum_{n=1}^{\infty} \mathcal{K}_n(t)$$

where

$$\begin{aligned} \mathcal{K}_n(t) = & \int_0^t dt_1 \int_0^{t_1} dt_2 \cdots \int_0^{t_{n-2}} dt_{n-1} \\ & \times \sum_q (-1)^q \mathcal{P} \mathcal{L}_t \cdots \mathcal{L}_{t_i} \mathcal{P} \mathcal{L}_{t_j} \cdots \mathcal{L}_{t_k} \mathcal{P} \cdots \mathcal{P} \end{aligned}$$

and where $\mathcal{L}_t = -i[H(t), t]$ and $H(t)$ is the Hamiltonian in rotating frame with respect to H_0 . The series is then approximated by stopping at a given order N , i.e. $\mathcal{K}_t \approx \sum_{n=1}^N \mathcal{K}_n(t)$.

Under these assumptions, a recursive formula similar to Eq. (14) has been derived. Specifically [9, Eq. (38)] (but also [10, Eq. (2)] [11, Eq. (2)], [12, Eq. (16)]) provides a recursive formula for the terms $\mathcal{K}_n(t)$ of the time-dependent generator $\mathcal{K}_t$. These recursive formulas however were derived in the rotating frame (w.r.t. H_0) under a weak coupling assumption and thus from an initial time-dependent model. This implies that the terms of the recursive formula contain time-ordered integrals that need to be solved in order to compute the terms of the expansion.

On the contrary, in this work, we start from a time-independent model, hence Eq. (14) (which was derived independently from [9–12]) does not contain integrals, thus resulting in overall simpler computations.

Furthermore, the main difference that separates this work from most of the physics literature, see e.g. [15, 33, 34] is the fact that instead of expanding the time-dependent generator in a power-series of a perturbation parameter ε here we expand in the time variable t . This difference allows for a few benefits: it simplifies the calculations of the approximations and lifts the weak-coupling assumption thus allowing to explore the effects of strong coupling regimes.

The main drawback introduced by the expansion in the time variable t is the fact that the approximate model holds only for small times. Nevertheless, as we highlighted above, while this consequence is clearly limiting, this situation is comparable to the typical derivation of TCL ME since no guarantees on the long-time limit are

available. On the contrary in this work we were able to characterize how the error between the original and reduced models grows with t .

C. Complete positivity of the first order

In this section we recall relevant known results about CPTP projectors, their images and reduced dynamics onto distorted algebras from the literature [13, 26–31]. The consequences of the assumptions we made about the projector $\mathcal{P}$ are summarized in the following proposition.

Proposition 1. *Let $\mathcal{P}$ be a CPTP projector with a full rank fixed point $\bar{\rho} > 0$, $\mathcal{P}(\bar{\rho}) = \bar{\rho}$ and let $\mathcal{V}$ be its image, i.e. $\mathcal{V} \equiv \text{Im} \mathcal{P} \subseteq \mathfrak{B}(\mathcal{H})$. Then there exist a decomposition of the Hilbert space*

$$\mathcal{H} = \bigoplus_k (\mathcal{H}_{F,k} \otimes \mathcal{H}_{G,k}),$$

a unitary operator $U \in \mathfrak{B}(\mathcal{H})$, and a set of density operators $\tau_k \in \mathfrak{D}(\mathcal{H}_{G,k})$ such that

$$\mathcal{V} = U \left(\bigoplus_k \mathfrak{B}(\mathcal{H}_{F,k}) \otimes \tau_k \right) U^\dagger. \quad (18)$$

Furthermore $\mathcal{V}$ is a $$ -algebra closed with respect to the product $X \cdot_\sigma Y \equiv X \sigma Y$ where*

$$\sigma = U \left(\bigoplus_k \mathbb{1}_{F,k} \otimes \tau_k^{-1} \right) U^\dagger.$$

Let then W_k be a linear operator from $\mathcal{H}$ onto $\mathcal{H}_{F,k} \otimes \mathcal{H}_{G,k}$ such that $W_k W_k^\dagger = \mathbb{1}_{F,k} \otimes \mathbb{1}_{G,k}$ [27]. Then the CPTP projection $\mathcal{P}$ onto $\mathcal{V}$ takes the form

$$\mathcal{P}(X) = U \left[\bigoplus_k \text{tr}_{\mathcal{H}_{G,k}} \left(W_k X W_k^\dagger \right) \otimes \tau_k \right] U^\dagger. \quad (19)$$

and is orthogonal with respect to the inner product $\langle X, Y \rangle_{\sigma, \lambda} = \text{tr}[X^\dagger \sigma^\lambda Y \sigma^{1-\lambda}]$ for $\lambda \in [0, 1]$.

Proof of this result can be found in [26] and [27, 28, 31, 35, 36]. Note that projectors of this type take the name of state extensions and are the dual of conditional expectations [24]. Furthermore, note that a $*$ -algebra closed with respect to a modified product often takes the name of *distorted algebra* [30], while the block structure provided in Eq.(18) takes the name of *Wedderburn decomposition*.

The previous result provides us with a structure for the image of a CPTP projection whenever there exists a full rank state $\bar{\rho}$ such that $\mathcal{P}(\bar{\rho}) = \bar{\rho}$. From this structure one can observe that a distorted algebra $\mathcal{V}$ is isomorphic to a smaller representation which we denote $\mathcal{A}$ where the weighted copies are removed, i.e. $\mathcal{A} \equiv \bigoplus_k \mathfrak{B}(\mathcal{H}_{F,k})$.

This allows us to find the two factors $\mathcal{R} : \mathfrak{B}(\mathcal{H}) \rightarrow \mathcal{A}$ and $\mathcal{J} : \mathcal{A} \rightarrow \mathfrak{B}(\mathcal{H})$ such that, $\mathcal{R}\mathcal{J} = \mathcal{I}_{\mathcal{A}}$ and $\mathcal{P} = \mathcal{J}\mathcal{R}$. Furthermore, one can observe [13] that these two factors can be chosen to be CPTP, i.e. for all $X \in \mathfrak{B}(\mathcal{H})$

$$\mathcal{R}(X) = \bigoplus_k \text{tr}_{\mathcal{H}_{G,k}}(W_k X W_k^\dagger) = \bigoplus_k X_{F,k} = \check{X}, \quad (20)$$

and for all $\check{X} \in \mathcal{A}$, $\check{X} = \bigoplus_k X_{F,k}$,

$$\mathcal{J}(\check{X}) = U \left(\bigoplus_k X_{F,k} \otimes \tau_k \right) U^\dagger. \quad (21)$$

The reduction and injection maps given in equations (20) and (21) then allow to compute the reduced dynamics $\mathcal{R}\mathcal{L}\mathcal{J}$ restricted onto the algebra $\mathcal{A}$. One can then prove that the reduced generator $\check{\mathcal{L}}$ is GKLS by leveraging the structure of the reduction and injection maps. We here report [13, Theorem 4] for completeness.

Theorem 3. *Let $\mathcal{P}$ be a CPTP projection with a full rank fixed point, let $\mathcal{V}$ be its image $\mathcal{V} = \text{Im}\mathcal{P} \subseteq \mathfrak{B}(\mathcal{H})$, and let $\mathcal{R}$ and $\mathcal{J}$ denote the CPTP factorization of $\mathcal{P} = \mathcal{J}\mathcal{R}$, as defined in Eq. (20) and (21). Then for any Lindblad generator $\mathcal{L}$, its restriction to $\mathcal{A}$, $\mathcal{R}\mathcal{L}\mathcal{J}$, is also a Lindblad generator, that is, $\mathcal{R}\mathcal{L}\mathcal{J} : \mathcal{A} \rightarrow \mathcal{A}$ and $\{e^{\check{\mathcal{L}}t}\}_{t \geq 0}$ is a quantum dynamical semi-group.*

Note that this theorem only proves the fact that $\mathcal{R}\mathcal{L}\mathcal{J}$ is of Lindblad type. In order to compute the reduced Hamiltonian and noise operators one can use the procedure described in [14, Appendix].

D. Second order: The case of purely Hamiltonian evolution

While the results we derived up to this point hold for general Lindblad generators $\mathcal{L}$, we shall now focus on the special case of purely Hamiltonian evolution, i.e. $\mathcal{L}(\cdot) = -i[H, \cdot]$. Other than the fact that this assumption is interesting in practical cases, we shall see that under this simplifying assumption we are able to prove that the second order term $\mathcal{F}_{(2)}$ is also of Lindblad-type.

The next result is based on the fact that the squared of a purely Hamiltonian generator is equal to a dissipator:

$$\begin{aligned} \mathcal{L}^2(\rho) &= -[H[H, \rho]] = -(H^2\rho - 2H\rho H + \rho H^2) \\ &= 2H\rho H - \{H^2, \rho\} = \mathcal{D}_{\sqrt{2}H}(\rho). \end{aligned} \quad (22)$$

This fact allows us to prove the next result.

Theorem 4. *Under the assumptions of Theorem 3 we have that $\mathcal{F}_{(2)} = \mathcal{R}\mathcal{L}^2\mathcal{J} - (\mathcal{R}\mathcal{L}\mathcal{J})^2$ is of Lindblad-type and for $\mathcal{F}_{t,2} \equiv \mathcal{F}_{(1)} + t\mathcal{F}_{(2)}$ we have that $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,2} ds}$ forms a set of CPTP maps for all $t \geq 0$.*

Proof. By leveraging [13, Proposition 5] we have that $\mathcal{R}\mathcal{L}\mathcal{J}(\check{\rho}) = -i[\check{H}, \check{\rho}]$ where $\check{H} = \mathcal{J}^\dagger(H)$ hence, using equation (22) we obtain $(\mathcal{R}\mathcal{L}\mathcal{J})^2(\check{\rho}) = \mathcal{D}_{\sqrt{2}\check{H}}(\check{\rho})$.

On the other hand, by leveraging [14, Proposition 6 and Proposition 7], we have that $\mathcal{R}\mathcal{D}_{\sqrt{2}H}\mathcal{J}(\check{\rho}) = \sum_{\check{C} \in \mathcal{X}(\sqrt{2}H)} \mathcal{D}_{\check{C}}(\check{\rho})$ for a set of operators $\mathcal{X}(\sqrt{2}H) \subset \mathfrak{B}(\mathcal{K})$. Importantly, by [14, Corollary 4], we have that $\mathcal{J}^\dagger(H) \in \mathcal{X}(\sqrt{2}H)$ and thus we can write $\mathcal{R}\mathcal{D}_{\sqrt{2}H}\mathcal{J}(\check{\rho}) = \sum_{\check{C} \in \mathcal{X}(\sqrt{2}H)/\{\sqrt{2}\check{H}\}} \mathcal{D}_{\check{C}}(\check{\rho}) + \mathcal{D}_{\sqrt{2}\check{H}}(\check{\rho})$ since $\sqrt{2}\check{H} = \mathcal{J}^\dagger(\sqrt{2}H)$. Then, by combining these two results we obtain

$$\mathcal{F}_{(2)} = \mathcal{R}\mathcal{L}^2\mathcal{J} - (\mathcal{R}\mathcal{L}\mathcal{J})^2 = \sum_{\check{C} \in \mathcal{X}(\sqrt{2}H)/\{\sqrt{2}\check{H}\}} \mathcal{D}_{\check{C}}(\check{\rho})$$

which is the sum of a set of GKLS, purely dissipative generators and thus of Lindblad type. Combining this with Theorem 3 it trivially follows that $\mathcal{F}_{t,2}$ is of Lindblad type for all $t \geq 0$. $\square$

E. Second order in the special case of bipartite systems

Because the proof of Theorem 4 is based on a few very technical results, we here want to present an pedagogical derivation of the same result in the special case of bipartite quantum systems. Note that this derivation is quite relevant in practice as it focuses on the most studied case of a bi-partite system but the cases included in the assumptions of Theorem 4 are more general. This exact case will be the focus of the two applications of interest studied in Sec V A and V B.

Let us assume that $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$ and let assume that the CPTP projector $\mathcal{P}$ is given by the $\mathcal{P}(\cdot) = \text{tr}_E(\cdot) \otimes \tau$ for some full rank state $\tau \in \mathfrak{D}(\mathcal{H}_E)$. The projector $\mathcal{P}$ is then factorized into the reduction map $\mathcal{R}(\cdot) = \text{tr}_E(\cdot)$ and the injection map $\mathcal{J}(\cdot) = \cdot \otimes \tau$. In line with the assumptions of this section we assume that $\mathcal{L}(\cdot) = -i\text{ad}_H(\cdot)$. Given the bipartite structure of the Hilbert space $\mathcal{H}$ we can always write $H = \sum_k S_k \otimes E_k$ for two sets of Hermitian operators $\{S_k\} \subset \mathfrak{B}(\mathcal{H}_S)$ and $\{E_k\} \subset \mathfrak{B}(\mathcal{H}_E)$.

We can then proceed to compute the reduced generator approximated at second order $\mathcal{F}_{t,2}(\mu)$. First, using [13, Proposition 5], we have that the generator at first order is

$$\mathcal{F}_{(1)}(\mu) = \mathcal{R}\mathcal{L}\mathcal{J}(\mu) = -i[\check{H}, \mu]$$

where

$$\check{H} = \mathcal{J}^\dagger(H) = \text{tr}_E[(\mathbb{1}_S \otimes \tau)H] = \sum_k a_k(\tau) S_k$$

and $a_k(\tau) \equiv \text{tr}[\tau E_k]$. Next, we can proceed to compute $\mathcal{F}_{(2)}$ by first computing $\mathcal{R}\mathcal{L}^2\mathcal{J}$:

$$\begin{aligned}
\mathcal{RL}^2 \mathcal{J}(\mu) &= \sum_{j,k} \text{tr}_E [2(S_j \otimes E_j)(\mu \otimes \tau)(S_k \otimes E_k) - (S_k \otimes E_k)(S_j \otimes E_j)(\mu \otimes \tau) - (\mu \otimes \tau)(S_k \otimes E_k)(S_j \otimes E_j)] \\
&= \sum_{j,k} 2S_j \mu S_k \underbrace{\text{tr}[E_j \tau E_k]}_{\equiv c_{j,k}(\tau)} - S_k S_j \mu \underbrace{\text{tr}[E_k E_j \tau]}_{=c_{j,k}(\tau)} - \mu S_k S_j \underbrace{\text{tr}[\tau E_k E_j]}_{=c_{j,k}(\tau)} \\
&= \sum_{j,k} c_{j,k}(\tau) [2S_j \mu S_k - \{S_k S_j, \mu\}].
\end{aligned}$$

On the other hand, since $(\mathcal{RL}\mathcal{J})^2 = \mathcal{D}_{\sqrt{2}\tilde{H}}$, we have

$$(\mathcal{RL}\mathcal{J})^2(\mu) = \sum_{j,k} a_j(\tau) a_k(\tau) [2S_j \mu S_k - \{S_k S_j, \mu\}]$$

and thus

$$\begin{aligned}
\mathcal{F}_{(2)}(\mu) &= [\mathcal{RL}^2 \mathcal{J} - (\mathcal{RL}\mathcal{J})^2](\mu) = \\
&= \sum_{j,k} [\chi(\tau)]_{j,k} [2S_j \mu S_k - \{S_k S_j, \mu\}]
\end{aligned}$$

where we defined the matrix $\chi(\tau)$ whose elements are $[\chi(\tau)]_{j,k} \equiv c_{j,k}(\tau) - a_j(\tau) a_k(\tau)$.

One can then notice that the matrix $\chi(\tau)$ coincides with a covariance matrix, i.e.

$$\begin{aligned}
[\chi(\tau)]_{j,k} &= \text{tr}[E_j \tau E_k] - \text{tr}[E_j \tau] \text{tr}[E_k \tau] \\
&= \mathbb{E}_\tau[(E_k - \mathbb{E}_\tau[E_k] \mathbb{1}_E)(E_j - \mathbb{E}_\tau[E_j] \mathbb{1}_E)]
\end{aligned}$$

hence $\chi(\tau)$ is a positive semi-definite matrix. Since $\chi(\tau)$ is positive semi-definite, it is diagonalizable with a unitary matrix, i.e. $UU^\dagger = U^\dagger U = \mathbb{1}$ such that $U\chi(\tau)U^\dagger = \gamma(\tau)$ where $\gamma(\tau)$ is a diagonal matrix of positive values (note that in reality U is also a function of τ). We can then define

$$L_h = \sum_j \overline{[U]_{h,j}} S_j$$

to obtain

$$\begin{aligned}
\mathcal{F}_{(2)}(\mu) &= \sum_{j,k} [\chi(\tau)]_{j,k} [2S_j \mu S_k - \{S_k S_j, \mu\}] \\
&= \sum_{j,k,h} \underbrace{[U^\dagger]_{j,h} [\gamma(\tau)]_h [U]_{h,k}}_{[\chi(\tau)]_{j,k}} [2S_j \mu S_k - \{S_k S_j, \mu\}] \\
&= \sum_{j,k,h} [\gamma(\tau)]_h \overline{[U]_{h,j}} [U]_{h,k} [2S_j \mu S_k - \{S_k S_j, \mu\}] \\
&= \sum_h [\gamma(\tau)]_h [2L_h \mu L_h^\dagger - \{L_h^\dagger L_h, \mu\}] \\
&= \sum_h [\gamma(\tau)]_h \mathcal{D}_{\sqrt{2}L_h}(\mu).
\end{aligned}$$

Because $\chi(\tau)$ is positive-semidefinite and thus $[\gamma(\tau)]_h$ are non-negative, then $\mathcal{F}_{(2)}$ is of Lindblad type. This implies

that the approximated time-dependent generator approximated at second order

$$\mathcal{F}_{t,2}(\mu) = -i[\tilde{H}, \mu] + \sum_h t[\gamma(\tau)]_h \mathcal{D}_{\sqrt{2}L_h}(\mu)$$

is also of Lindblad type at all times $t \geq 0$ as proved in Theorem 4.

Note that in deriving the reduced generator $\mathcal{F}_{t,2}$ there was no need to move the system's description into rotating frame. This provides another benefit of the proposed approach.

V. APPLICATIONS TO QUANTUM MODELS

In this section we consider a few examples of interest:

I) The first example considers a qubit coupled to a bosonic bath. As the exact solution of such a model is well known, this example allows us to assess the goodness of the approximation proposed in this work.

II) Second, a qubit coupled to a finite bath is studied. Given the fact that the bath is finite-dimensional this is typically considered an “highly non-Markovian model”. This example allows us to showcase the numerical procedure at higher orders for both a case where the original dynamics is purely Hamiltonian and a case where a dissipative term is included.

III) Lastly an Ising spin chain is studied. This example allows us to showcase that the proposed procedure works in cases that are more general than the bipartite case and allows us to show that a non-Markovian reduced model can be obtained also for the population of the state (and not only for the bipartite case).

A. Dephasing spin boson model

Inspired by [5, 37–39], in this section we consider a dephasing spin-boson model to further showcase the advances and drawbacks of the proposed method. Such a model is typically used to model the dephasing of qubits in quantum computers [40]. Note that: i) in this section we make an exception to the assumption of final dimensional systems considered throughout the paper; ii) the model we are going to consider is exactly solvable hence

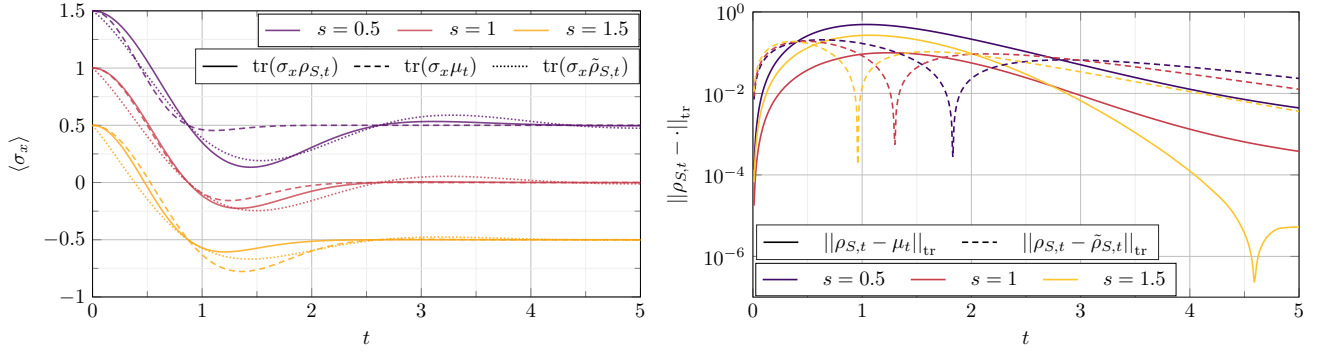


Figure 4: Simulation of the dephasing spin-boson model for $s = 0.5, 1, 1.5$ where we compare the exact solution $\rho_{S,t}$ given in Eq.(26), the second order approximation μ_t derived in this work, given in Eq. (27) and the coarse-grained dynamics $\tilde{\rho}_{S,t}$ given in Eq. (28). Left: Evolution of the expectation value $\langle \sigma_x \rangle$ for the exact solution (continuous curves), second order approximation (dashed curve) and coarse-grained dynamics (dotted curve). The expectations values of the cases $s = 0.5, 1.5$ have been shifted by ± 0.5 respectively for graphical purposes. Right: evolution of the trace-norm error with respect to the exact solution committed by the second order approximation (continuous curve) and coarse-grained dynamics (dashed curve).

the intent of this subsection is to showcase that the proposed approximation method provides results comparable to other approximation techniques, not to compare it against the exact solution of the model. Nevertheless, this example helps us to understand the degrees of approximation that the proposed method offers and how the process unfolds.

The model is composed of a spin $\frac{1}{2}$ system $\mathcal{H}_S \simeq \mathbb{C}^2$ coupled to a bath $\mathcal{H}_B$ composed of N_B bosonic modes, $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$. The model's dynamics is driven by the total Hamiltonian $H = H_S \otimes \mathbb{1}_B + \mathbb{1}_S \otimes H_B + H_{\text{int}}$ where

$$H_S = \frac{g}{2} \sigma_z \quad (23)$$

$$H_B = \sum_{k=1}^{N_B} \omega_k \left(n_k + \frac{1}{2} \right) \quad (24)$$

$$H_{\text{int}} = \sigma_z \otimes \left(\sum_{k=1}^{N_B} \lambda_k b_k + \bar{\lambda}_k b_k^\dagger \right) \quad (25)$$

and where $n_k = b_k^\dagger b_k$ and b_k are the bosonic number and annihilation operator for mode k respectively and satisfy $[b_k, b_l^\dagger] = \delta_{k,l} \mathbb{1}$, $\omega_k \in \mathbb{R}$ is the mode frequency and $\lambda_k \in \mathbb{C}$ is the strength of the coupling between the mode and the spin.

We assume the initial state to be factorized $\rho_0 = \rho_{S,0} \otimes \rho_B$ with $\rho_B = e^{-\beta H_B} / \text{tr}[e^{-\beta H_B}]$ and some state $\rho_{S,0} \in \mathcal{D}(\mathcal{H}_S)$ for the spin. We further assume the projector $\mathcal{P}$ to be $\mathcal{P}(\rho) = \text{tr}_B(\rho) \otimes \rho_B$ which can be factorized into $\mathcal{R}(\rho) = \text{tr}_B(\rho)$ and $\mathcal{J}(\sigma) = \sigma \otimes \rho_B$. One can then verify that $\mathcal{P}(\rho_0) = \rho_0$.

This model has a known exact analytical solutions given by [39, eq. (336) and (374)]:

$$\rho'_{S,t} = \begin{bmatrix} [\rho_{S,0}]_{0,0} & e^{-2\gamma(t)t} [\rho_{S,0}]_{0,1} \\ e^{-2\gamma(t)t} [\rho_{S,0}]_{1,0} & [\rho_{S,0}]_{1,1} \end{bmatrix}$$

where $\rho'_{S,t}$ is the spin's state in the rotating frame and $\gamma(t) = \sum_k \alpha_k \frac{1 - \cos(\omega_k t)}{t}$ with $\alpha_k \equiv 2 \frac{|\lambda_k|^2}{\omega_k^2} \coth\left(\frac{\beta \omega_k}{2}\right) \in \mathbb{R}$. By taking the derivative of an off diagonal element we obtain:

$$\frac{d}{dt} [\rho'_{S,t}]_{0,1} = -2 \underbrace{\left[\gamma(t) + \frac{d\gamma(t)}{dt} t \right]}_{\equiv \xi(t)} [\rho'_{S,t}]_{0,1}$$

where $\xi(t) = \sum_k \alpha_k \omega_k \sin(\omega_k t)$. This dynamics corresponds to the differential equation $\frac{d}{dt} \rho'_{S,t} = \xi(t) \mathcal{D}_{\sigma_z} [\rho'_{S,t}]$ for the state in rotating frame or, equivalently, to

$$\dot{\rho}_{S,t} = -i[H_S, \rho_{S,t}] + \xi(t) \mathcal{D}_{\sigma_z}(\rho_{S,t}) \quad (26)$$

for the state in the lab frame.

By following the procedure detailed in Sec. IVE we can then proceed to compute the reduced time-dependent generator at second order $\mathcal{F}_{t,2}$. Denoting $\langle X \rangle_B \equiv \text{tr}[\rho_B X]$, we recall that [39, eq. (331)]:

$$\begin{aligned} \langle b_k^\dagger \rangle_B &= \langle b_k \rangle_B = \langle b_k^\dagger b_l^\dagger \rangle_B = \langle b_k b_l \rangle_B = 0, \\ \langle b_j^\dagger b_k \rangle_B &= \frac{\delta_{j,k}}{e^{\beta \omega_k} - 1}. \end{aligned}$$

We first compute the reduced Hamiltonian $\tilde{H} = \mathcal{J}^\dagger(H)$ where $\mathcal{J}^\dagger(X) = \text{tr}_B[X(\mathbb{1}_S \otimes \rho_B)]$. By noting that $\text{tr}_B[H_S \otimes \mathbb{1}_B(\mathbb{1}_S \otimes \rho_B)] = H_S \text{tr}[\rho_B] = H_S$, $\text{tr}_B[\mathbb{1}_S \otimes H_B(\mathbb{1}_S \otimes \rho_B)] = \mathbb{1}_S \langle H_B \rangle_B$ and

$$\begin{aligned} \text{tr}_B \left[\sigma_z \otimes \left(\sum_{k=1}^{N_B} \lambda_k b_k + \bar{\lambda}_k b_k^\dagger \right) (\mathbb{1}_S \otimes \rho_B) \right] \\ = \sigma_z \left(\sum_{k=1}^{N_B} \lambda_k \langle b_k \rangle_B + \bar{\lambda}_k \langle b_k^\dagger \rangle_B \right) = 0 \end{aligned}$$

we have $\check{H} = H_S$ as the term $\mathbb{1} \langle H_B \rangle_B$ only contributes to a phase and can thus be removed leading to the first order term $\mathcal{F}_{(1)}(\mu) = -i[\check{H}, \mu]$. The second order term is instead found to be $\mathcal{F}_{(2)}(\mu) = \varphi \mathcal{D}_{\sigma_z}(\mu)$ with $\varphi \equiv \sum_{k=1}^{N_B} \frac{2|\lambda_k|^2}{e^{\beta\omega_k}-1}$. All the necessary calculations are detailed in Appendix B for the sake of clarity.

Combining both terms we obtain an approximate reduced model described by the time-dependent Lindblad equation

$$\dot{\mu}_t = -i[H_S, \mu_t] + \varphi t \mathcal{D}_{\sigma_z}(\mu_t). \quad (27)$$

By comparing this equation with the exact solution of Eq. (26) we can observe that both the Hamiltonian and noise operator are correctly recovered whereas the time-dependent dissipation strength $\xi(t)$ is approximated with φt . Interestingly, the value φ does not coincide with the first order Taylor expansion of $\xi(t)$ which is $\sum_k \alpha_k \omega_k$.

To better assess the quality of the reduced second order approximation we next compare the evolution of $\rho_{S,t}$ and that of μ_t with the coarse-grained approximation discussed in [39, eq. (332)], i.e. $\tilde{\rho}_{S,t} \in \mathfrak{D}(\mathcal{H}_S)$ and

$$\dot{\tilde{\rho}}_{S,t} = -i[H_S, \tilde{\rho}_{S,t}] + \gamma(\tau) \mathcal{D}_{\sigma_s}(\tilde{\rho}_{S,t}) \quad (28)$$

where the constant $\tau \in \mathbb{R}$ is specified later.

The bath coupling coefficients are going to be defined as $\lambda_k = \sqrt{J(\omega_k)}$ [41] where $J(\omega)$ is the bath spectral density and is chosen to be

$$J(\omega) = \Lambda \omega_c^{1-s} \omega^s e^{-\frac{\omega}{\omega_c}} \quad (29)$$

where $\Lambda \in \mathbb{R}_+$ is an overall constant, $\omega_c \in \mathbb{R}_+$ is the bath cutoff frequency and $s \in \mathbb{R}_+$ is the Ohmicity parameter which determines three cases of interest, i.e. Ohmic ($s = 1$), sub-Ohmic ($s < 1$) and super-Ohmic ($s > 1$) [42].

In Fig. 4 the results of the simulations are shown in the three cases, i.e. $s = 0.5, 1, 1.5$. The other parameters used in the simulations are $g = 1.8$, $\omega_c = 0.2$, $\Lambda = 0.5$, $\beta = 10$, $\tau = 32/\omega_c$, and where ω_k are $N_B = 100$, taken equally spaced between 0 and 1. The initial state instead has been fixed to $\rho_{S,0} = \mu_0 = \tilde{\rho}_{S,0} = |+\rangle\langle+|$. Importantly, one can observe in Fig. 4 Right that, in each of the three cases, the error committed by the second order approximation, Eq. (27) is below that of the coarse-grained dynamics, Eq. (28) for both the initial and final phases while it is above for a central phase. This indicates that the second order approximation performs better than the coarse-grained Markovian approximation in this case.

B. Dissipative central spin model

In this section we consider a dissipative central spin model to showcase the capabilities of the proposed technique. Similar models have been studied in the literature, see e.g. [43–46]. The model is composed by a system, i.e. $\mathcal{H}_S \simeq \mathbb{C}^2$ also denoted as spin 1 and a bath composed

of N_B other spins, i.e. $\mathcal{H}_B \simeq \mathbb{C}^{2^{N_B}}$ counted from 2 to $N_B + 1$.

The system Hamiltonian is described as $H = H_S + H_B + H_{\text{int}}$ where:

$$\begin{aligned} H_S &= \delta(\sigma_x^{(1)} + \sigma_z^{(1)}) \\ H_B &= \frac{\gamma}{4} \left(2J_x^2 - \frac{N_B}{2} \mathbb{1} \right) \\ H_{\text{int}} &= \frac{\lambda}{2} \left(A_x \sigma_x^{(1)} J_x + A_y \sigma_y^{(1)} J_y + A_z \sigma_z^{(1)} J_z \right) \end{aligned}$$

where $\sigma_u^{(j)}$ denotes the $u \in \{x, y, z\}$ Pauli matrix acting on the j -th spin and $J_u = \frac{1}{2} \sum_{k=2}^{N_B+1} \sigma_u^{(k)}$ denotes the total bath angular momentum operators. The bath is also in contact with a Markovian environment that dissipates information. This interaction is described by the noise operators $L_k = \Lambda \sigma_-^{(k)}$ for all $k = 2, \dots, N_B + 1$ where $\sigma_{\pm}^{(k)} = \frac{1}{2}(\sigma_x^{(k)} \pm \sigma_y^{(k)})$.

For this example we consider as an initial condition the factorized state $\rho_0 = |+\rangle\langle+| \otimes \rho_B$ where

$$\rho_B = \frac{e^{-\beta H_B}}{\text{tr}[e^{-\beta H_B}]}$$

is the thermal state at inverse temperature β . This allows us to define the reduction and injection maps as $\mathcal{R}(\cdot) = \text{tr}_B(\cdot)$ and $\mathcal{J}(\cdot) = \cdot \otimes \rho_B$. Note that with this choice we have that $\mathcal{P}(\rho_0) = \rho_0$. The reduced state then corresponds to the state of the central spin, i.e.

$$\eta_t = \rho_S(t) \equiv \text{tr}_B(\rho_t).$$

In Fig. 5 the results of the simulation for this dissipative central spin model are shown. The simulation has been run in two settings: a purely Hamiltonian setting ($\Lambda = 0$ on the left) and a dissipative setting ($\Lambda = 0.8$ on the right). The rest of the parameters used were $N_B = 3$, $\delta = 0.3$, $\lambda = 0.1$, $\gamma = 1$, $A_x = 1.2$, $A_y = 1.5$, $A_z = 1.3$ and $\beta = 50$. In the numerical simulations we can observe a few interesting aspects: 1) as expected, the error between the exact solution and the approximate solution $\|\eta_t - \mu_t\|_t$ decreases when increasing the approximation degree N ; 2) the trajectory for approximation orders $N = 1, 2$ and even 3 do not exit the Bloch sphere (note that we have proven the complete positivity for orders $N = 1, 2$ but not for $N = 3$); 3) the simulations for $N > 3$ are displaying Runge's phenomena at larger times, which is quite typical in these situations; 4) after some time, the evolved state η_t for an approximation degree $N \geq 4$ exits the Bloch sphere, indicating that the dynamics is not CPTP at all times for approximation degrees higher than $N \geq 4$.

C. Ising spin chain

While the idea of approximating a large model with a smaller one that is LTV is common in the setting of a

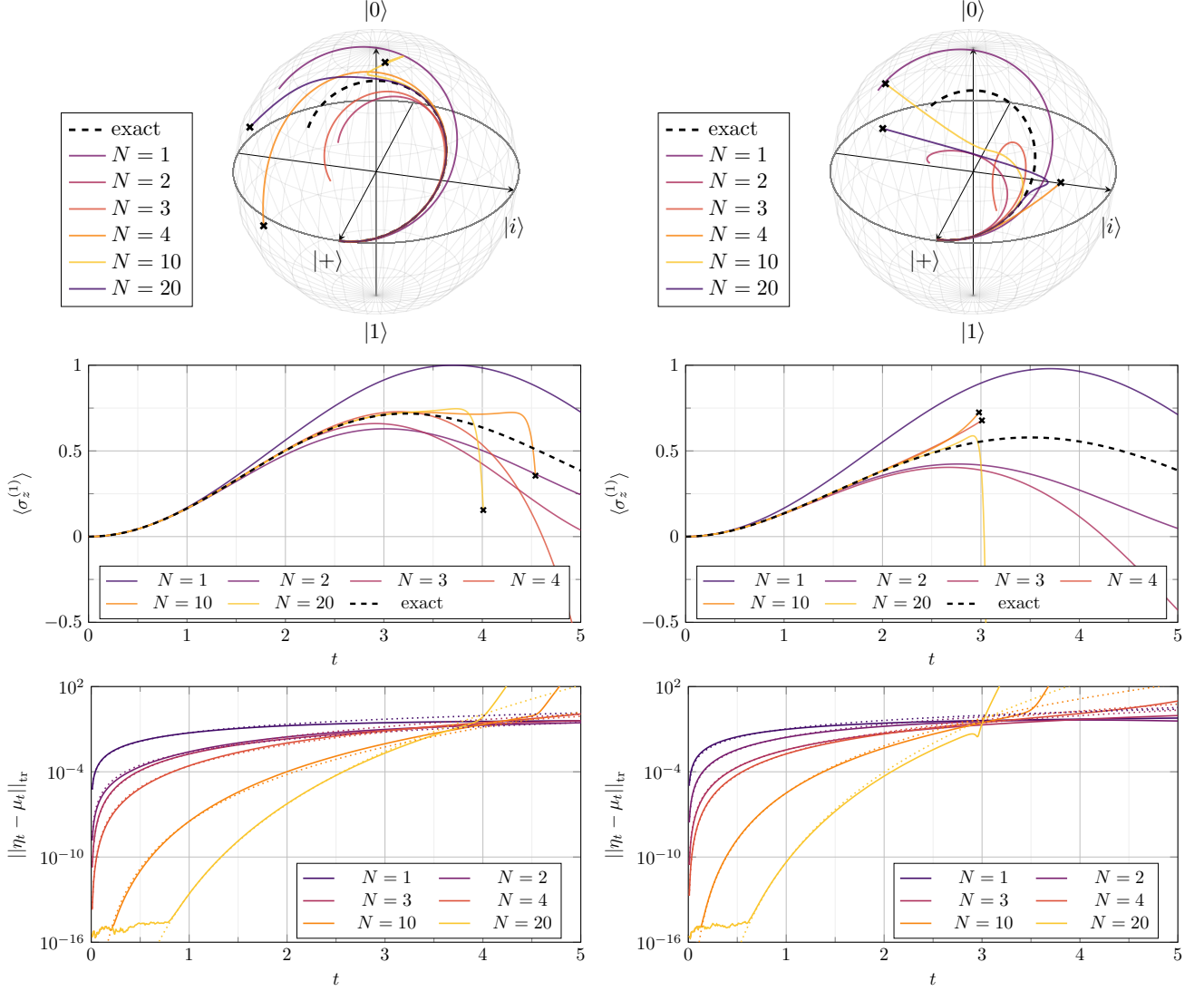


Figure 5: Simulations of the dissipative central spin model in the purely Hamiltonian setting ($\Lambda = 0$) on the left and in the dissipative setting ($\Lambda = 0.8$) on the right. The exact evolution and various degrees of approximations ($N = 1, 2, 3, 4, 10, 20$) are shown in each figure. The first row shows the trajectory of the central spin's state ρ_S in the Bloch sphere; the second row shows the magnetization of the central spin in the σ_z direction versus time; the third row shows the trace norm error between the exact trajectory η_t and the reduced one μ_t , i.e. $\|\eta_t - \mu_t\|_{\text{tr}}$ versus time for different values of the approximation N and with dotted curve representing a fitted αt^{N+1} curve. The black crosses denote points where the evolved state μ_t exits the set of density operators.

bipartite system-environment couple, this is far from the only case where such an approach is useful. More in general, the same approaches can be adapted to cases where one wants to describe the evolution of only a small set of variables of interest of the original model. For this reason in this section we showcase the use of using the same methods presented before to reproduce the probability distribution in the standard basis.

In order to show that the proposed method works also in cases that are not bi-partite we here consider a spin-chain model. More specifically we consider a model composed of N_B spins, i.e. $\mathcal{H} \simeq \mathbb{C}^n$ where $n = 2^{N_B}$ whose

dynamics is driven by the Ising Hamiltonian

$$H = \sum_{j=1}^{N_B} h \sigma_z^{(j)} + A \sum_{j=1}^{N_B-1} \sigma_x^{(j)} \sigma_x^{(j+1)}. \quad (30)$$

In this example we consider $\mathcal{P}$ to be the CPTP projector onto the diagonal $\mathcal{P}(\rho) = \sum_{j=0}^{n-1} |j\rangle\langle j| \rho |j\rangle\langle j|$ which can be factorized into $\mathcal{R}(\rho) = \sum_{j=0}^{n-1} |j\rangle\langle j| \rho |j\rangle\langle j|$ and $\mathcal{J}(\eta) = \sum_{j=0}^{n-1} |j\rangle\langle j| \langle j| \eta$. Note that, in such a case, $\eta = \mathcal{R}(\rho) \in \mathbb{R}^n$ is a probability distribution, i.e. $\eta_i \geq 0$ and $\mathbf{1}^T \eta = 1$ where $\mathbf{1} \in \mathbb{R}^n$ is the vector of all ones. This implies that $\mathcal{F}_t \in \mathbb{R}^{n \times n}$.

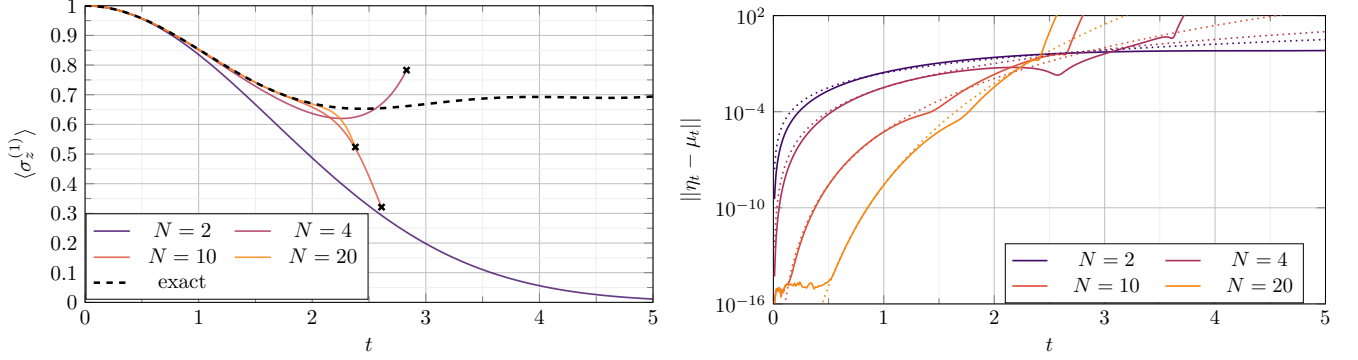


Figure 6: Simulation of the Ising spin chain. Magnetization of the first spin of the chain $\langle \sigma_z^{(1)} \rangle$ versus time on the left and the error norm $\|\eta_t - \mu_t\|_2$ versus time on the right. Various approximation's degrees $N = 2, 4, 10, 20$ are shown in each figure. The black crosses denote the points where the evolved state μ_t exits the n -dimensional simplex, i.e. μ_t is no longer a probability distribution while the dotted curves on the right again represent fitted curves of αt^{N+1} .

As an initial condition we consider $\rho_0 = |0\rangle\langle 0|$. Trivially, $\mathcal{P}(\rho_0) = \rho_0$.

Numerically, we verified that $\mathcal{F}_{(1)} = \mathcal{F}_{(3)} = 0$ while $\mathcal{F}_{(2)}$ is graphically represented in Fig. 7 where one can verify that $\mathcal{F}_{(2)}$ is Metzler (a matrix A is Metzler if $A_{i,j} \geq 0$ for all $i \neq j$) and is such that $\mathbf{1}^T \mathcal{F}_{(2)} = 0$. Those two conditions are the classical equivalent to the properties characterizing a Lindblad generator and in fact imply that $e^{\mathcal{F}_{(2)}t}$ is a stochastic matrix for all $t \geq 0$. This directly implies that $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,2} ds}$ is CPTP, or stochastic in this case, for all $t \geq 0$.

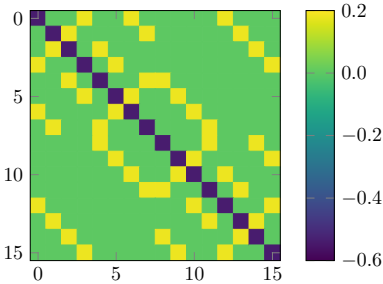


Figure 7: Graphical representation of the term $\mathcal{F}_{(2)}$ for the time-dependent generator $\mathcal{F}_t$ of the Ising spin chain model.

Fig. 6 shows the results of the simulations of the Ising spin chain model with $N_B = 4$, $A = 0.3$, $h = 0.36$ where we can observe similar results to the other simulations included in this work: increasing the approximation degree decreases the error $\|\eta_t - \mu_t\|_2$ at low times; for $N = 2$ the approximate evolved state μ_t remains a probability distribution while for $N \geq 4$, after a certain time, the state is no longer a probability distribution.

VI. CONCLUSIONS AND OUTLOOK

In this work we explored how model reduction can benefit from a reduced model that is time-dependent. The problem of finding time dependent reduced model is motivated by the derivation of time-convolutionless master equations in quantum mechanics but can in principle be applied to any linear model.

The main contribution of this work consists of a novel method to compute a time-dependent model that is polynomial in time and that approximate the evolution of the original model well for small times. Importantly, the coefficients of the polynomial can be determined using a recursive formula, Eq (14) that is far simpler than the alternatives. This method thus offer an alternative to the standard perturbative derivation of TCL ME.

Benefits of the proposed reduction method include: the method does not require to move into a rotating frame, making the calculations simpler; the iterative form of the coefficients allow for easy calculations of the terms and, as shown in Sec. III A, for lower orders the method offers an advantage with respect to truncating the exponential; the proposed method applies to any linear model (not only purely Hamiltonian dynamics); the method does not rely on a perturbation on some supposedly small parameter ε allowing for analysis of systems into stronger coupling regimes; when the original dynamics is purely Hamiltonian, the reduced time-dependent model is ensured to generate a CPTP two-parameters semigroup at all times. All of this comes at the cost of having a model that approximates the original evolution in the short-time regime.

Future work shall focus on: i) extending Theorem 1 and Theorem 2 to the case of analytic generators F_t (whose main challenge is described in Appendix A); ii) Lift the assumption $Px_0 = x_0$; iii) Apply the proposed procedure to other quantum models in the strong coupling regime to study models that were not accessible

before and further test the capabilities of the method; iv) combine the classical model reduction methods and the time-dependent model reduction proposed here to see if the combination of the two methods can allow to improve the accuracy of the reduced model. Note that a first step in this direction has been taken in the physics literature by [19, 20] where, including extra degrees of freedom, has been shown to improve the accuracy of the derived TCL ME.

Given the numerical examples of the central spin model presented in Sec. VB we do not expect the reduced model to retain the CP property for orders higher than $N = 3$. It is however possible that, in the Hamiltonian case, for $N = 3$ one can prove that the evolution of the reduced model is necessarily CP at all times as none of the simulation carried out while preparing this work contradicted this statement.

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Code availability: The code necessary to reproduce all the numerical experiments presented in this work is available at [47].

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Appendix A: Technical results and proofs for the time-ordered expansion

Before proving the main result, Theorem 1, we shall develop some intuition about how we prove it.

Example 1. To provide some intuition about Theorem 1 let us consider a simple case: $F_t = F_{(1)} + tF_{(2)} + t^2F_{(3)}$. Let us first compute the first few terms of the Dyson expansion [48] of $\mathbb{T}e^{\int_0^t F_s ds}$, i.e. $\mathbb{T}e^{\int_0^t F_s ds} = \sum_{n=0}^{+\infty} \Phi_{(n),t}$. At zeroth order we have $\Phi_{(0),t} = \mathcal{I}$ while at orders 1, 2 and 3 we have:

$$\Phi_{(1),t} = \int_0^t F_s ds = \int_0^t (F_{(1)} + sF_{(2)} + s^2F_{(3)})ds = tF_{(1)} + t^2\frac{F_{(2)}}{2} + t^3\frac{F_{(3)}}{3},$$

$$\begin{aligned}
\Phi_{(2),t} &= \int_0^t dt_1 \int_0^{t_1} dt_2 F_{t_1} F_{t_2} = \sum_{j=0}^1 \int_0^t dt_1 F_{t_1} F_{(j+1)} \int_0^{t_1} dt_2 t_2^j = \sum_{j=0}^1 \int_0^t dt_1 F_{t_1} F_{(j+1)} \frac{t_1^{j+1}}{j+1} \\
&= \sum_{k,j=0}^1 F_{(k+1)} F_{(j+1)} \int_0^t dt_1 t_1^k \frac{t_1^{j+1}}{j+1} = \sum_{k,j=0}^1 F_{(k+1)} F_{(j+1)} \frac{t^{k+j+2}}{(k+j+2)(j+1)} = \sum_{k,j=1}^2 F_{(k)} F_{(j)} \frac{t^{k+j}}{(j+k)j} \\
&= \underbrace{F_{(1)}^2 \frac{t^2}{2}}_{j=k=1} + \underbrace{F_{(2)} F_{(1)} \frac{t^3}{3}}_{j=1,k=2} + \underbrace{F_{(1)} F_{(2)} \frac{t^3}{6}}_{j=2,k=1} + O(t^4), \\
\Phi_{(3),t} &= \int_0^t dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 F_{t_1} F_{t_2} F_{t_3} = \int_0^t dt_1 F_{t_1} \underbrace{\int_0^{t_1} dt_2 \int_0^{t_2} dt_3 F_{t_2} F_{t_3}}_{=\sum_{j,k=1}^2 F_{(k)} F_{(j)} \frac{t_1^{k+j}}{(j+k)j}} \\
&= \sum_{h,k,j=1}^2 F_{(h)} F_{(k)} F_{(j)} \int_0^t dt_1 t_1^h \frac{t_1^{k+j}}{(j+k)j} = \sum_{h,k,j=1}^2 F_{(h)} F_{(k)} F_{(j)} \frac{t^{h+k+j}}{(h+k+j)(j+k)j} = \underbrace{F_{(1)}^3 \frac{t^3}{3!}}_{h=k=j=1} + O(t^4).
\end{aligned}$$

Grouping together the terms of $\mathbb{T}e^{\int_0^t F_s ds}$ by the order of the exponent of t , we have

$$\mathbb{T}e^{\int_0^t F_s ds} = \underbrace{\mathcal{I}}_{E_{(0)}} + t \underbrace{F_{(1)}}_{E_{(1)}} + t^2 \underbrace{\left(\frac{F_{(2)}}{2} + \frac{F_{(1)}^2}{2} \right)}_{E_{(2)}} + t^3 \underbrace{\left(\frac{F_{(3)}}{3} + \frac{F_{(1)} F_{(2)}}{6} + \frac{F_{(2)} F_{(1)}}{3} + \frac{F_{(1)}^3}{3!} \right)}_{E_{(3)}} + O(t^4)$$

which coincides with the first few terms obtained using the equation given in Theorem 1:

$$\begin{aligned}
E_{(0)} &= \mathcal{I}, \\
E_{(1)} &= F_{(1)} E_{(0)} = F_{(1)}, \\
E_{(2)} &= \frac{1}{2} (F_{(1)} E_{(1)} + F_{(2)} E_{(0)}) = \frac{1}{2} (F_{(1)}^2 + F_{(2)}), \\
E_{(3)} &= \frac{1}{3} (F_{(1)} E_{(2)} + F_{(2)} E_{(1)} + F_{(3)} E_{(0)}) = \frac{1}{3} \left(\frac{1}{2} F_{(1)}^3 + \frac{1}{2} F_{(1)} F_{(2)} + F_{(2)} F_{(1)} + F_{(3)} \right).
\end{aligned}$$

□

Theorem 1. We shall start by proving that $\mathbb{T}e^{\int_0^t F_s ds} = \sum_{n=0}^{+\infty} \Phi_{(n),t}$ where $\Phi_{(0),t} = \mathcal{I}$ and, for $n > 0$,

$$\Phi_{(n),t} = \sum_{k_1, \dots, k_n=1}^{N+1} t^{\sum_{j=1}^n k_j} \frac{F_{(k_1)} \dots F_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}.$$

First of all, using the Dyson series expansion for the time-ordered exponential $\mathbb{T}e^{\int_0^t F_s ds}$, see e.g. [48–50], we have that:

$$\mathbb{T}e^{\int_0^t F_s ds} = \sum_{n=0}^{+\infty} \Phi_{(n),t} \quad \text{with} \quad \Phi_{(n),t} = \int_0^t dt_1 \int_0^{t_1} dt_2 \dots \int_0^{t_{n-1}} dt_n F_{t_1} F_{t_2} \dots F_{t_n}.$$

Next, by noticing that $\Phi_{(n),t} = \int_0^t ds F_s \Phi_{(n-1),s}$ we can prove the statement by induction. The case $n = 1$ is proven by simple calculations:

$$\Phi_{(1),t} = \int_0^t ds F_s = \int_0^t ds \sum_{k_1=0}^N s^{k_1} F_{(k_1+1)} = \sum_{k_1=0}^N F_{(k_1+1)} \int_0^t ds s^{k_1} = \sum_{k_1=0}^N F_{(k_1+1)} \frac{t^{k_1+1}}{k_1+1} = \sum_{k_1=1}^{N+1} F_{(k_1)} \frac{t^{k_1}}{k_1}.$$

Note that $\int_0^t ds \sum_{k=0}^N s^k = \sum_{k=0}^N \int_0^t ds s^k$ under the assumption that N is finite, i.e. $N < \infty$. We next assume that the statement holds for n , i.e.

$$\Phi_{(n),t} = \sum_{k_1, \dots, k_n=1}^{N+1} t^{\sum_{j=1}^n k_j} \frac{F_{(k_1)} \dots F_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}.$$

Then

$$\begin{aligned}
\Phi_{(n+1),t} &= \int_0^t ds F_s \Phi_{(n),s} = \int_0^t ds F_s \sum_{k_1, \dots, k_n=1}^{N+1} s^{\sum_{j=1}^n k_j} \frac{F_{(k_n)} \dots F_{(k_1)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\
&= \int_0^t ds \sum_{k=0}^N s^k F_{(k+1)} \sum_{k_1, \dots, k_n=1}^{N+1} s^{\sum_{j=1}^n k_j} \frac{F_{(k_1)} \dots F_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\
&= \sum_{k=0}^N \sum_{k_1, \dots, k_n=1}^{N+1} F_{(k+1)} \frac{F_{(k_1)} \dots F_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \int_0^t s^{k+\sum_{j=1}^n k_j} ds \\
&= \sum_{k=0}^N \sum_{k_1, \dots, k_n=1}^{N+1} \frac{t^{k+1+\sum_{j=1}^n k_j}}{k+1+\sum_{j=1}^n k_j} F_{(k+1)} \frac{F_{(k_1)} \dots F_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\
\Phi_{(n+1),t} &= \sum_{k=1}^{N+1} \sum_{k_1, \dots, k_n=1}^{N+1} \frac{t^{k+\sum_{j=1}^n k_j}}{k+\sum_{j=1}^n k_j} F_{(k)} \frac{F_{(k_1)} \dots F_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} = \sum_{k_1=1}^{N+1} \sum_{k_2, \dots, k_{n+1}=1}^{N+1} \frac{t^{k_1+\sum_{j=2}^{n+1} k_j}}{k_1+\sum_{j=2}^{n+1} k_j} \frac{F_{(k_1)} F_{(k_2)} \dots F_{(k_{n+1})}}{\prod_{j=1}^n \sum_{h=n-j+2}^{n+1} k_h} \\
&= \sum_{k_1, \dots, k_{n+1}=1}^{N+1} \frac{t^{\sum_{j=1}^{n+1} k_j}}{\sum_{j=1}^{n+1} k_j} \frac{F_{(k_1)} \dots F_{(k_{n+1})}}{\prod_{j=1}^{n+1} \sum_{h=n-j+2}^{n+1} k_h} = \sum_{k_1, \dots, k_{n+1}=1}^{N+1} t^{\sum_{j=1}^{n+1} k_j} \frac{F_{(k_1)} \dots F_{(k_{n+1})}}{\prod_{j=1}^{n+1} \sum_{h=n-j+2}^{n+1} k_h}.
\end{aligned}$$

thus proving the statement.

We next prove that $\mathbb{T}e^{\int_0^t F_s ds} = \sum_{k=0}^{+\infty} t^k E_{(k)}$ where $E_{(0)} = \mathcal{I}$ and, for $k > 0$,

$$E_{(k)} = \sum_{n=1}^k \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k}}^k \frac{F_{(k_1)} \dots F_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}.$$

To prove this claim we can start by noticing that each term $\Phi_{(n),t}$ contains powers t^k of order $k \geq n$. Hence, to construct $E_{(k)}$ of order k we need to collect and sum all the coefficients that multiply t^k among the terms $\Phi_{(n),t}$ for $1 \leq n \leq k$. Given $\Phi_{(n),t}$ with $n \leq k$ we have that the coefficient multiplying t^k must satisfy the constraint $\sum_{j=1}^n k_j = k$. Thus, the coefficient of $\Phi_{(n),t}$ that multiply t^k are

$$\sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k}}^{N+1} \frac{F_{(k_1)} \dots F_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} = \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k}}^k \frac{F_{(k_1)} \dots F_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}$$

where we used the fact that, if any $k_j > k$ then $\sum_{j=1}^n k_j > k$. Noticing that $\prod_{j=1}^n \sum_{h=n-j+1}^n k_h = k(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)$ and summing over $1 \leq n \leq k$ we obtain, for $k > 0$, the second claim, i.e.

$$E_{(k)} = \sum_{n=1}^k \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k}}^k \frac{F_{(k_1)} \dots F_{(k_n)}}{k(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)}.$$

We are now finally ready to prove the main claim of the theorem, that is: $E_{(k)} = \frac{1}{k} \sum_{s=1}^k F_{(s)} E_{(k-s)}$. Recalling that $E_{(0)} = \mathcal{I}$ we can write:

$$\begin{aligned}
\frac{1}{k} \sum_{s=1}^k F_{(s)} E_{(k-s)} &= \sum_{s=1}^{k-1} \frac{F_{(s)}}{k} E_{(k-s)} + \frac{F_{(k)}}{k} \\
&= \sum_{s=1}^{k-1} \sum_{n=1}^{k-s} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k-s}}^k \frac{F_{(s)}}{k} \frac{F_{(k_1)} \dots F_{(k_n)}}{(k-s)(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{F_{(k)}}{k} \\
&= \sum_{s=1}^{k-1} \sum_{n=1}^{k-1} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k-s}}^k \frac{F_{(s)}}{k} \frac{F_{(k_1)} \dots F_{(k_n)}}{(k-s)(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{F_{(k)}}{k}
\end{aligned}$$

where we used the fact that $\sum_{n=k-s+1}^{k-1} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k-s}}^k \frac{F(s)}{k} \frac{F(k_1) \dots F(k_n)}{(k-s)(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} = 0$ since for $n > k-s$ we have that $\sum_{j=1}^n k_j > k-s$ for any set of k_j . This implies that

$$\begin{aligned}
\frac{1}{k} \sum_{s=1}^k F(s) E_{(k-s)} &= \sum_{n=1}^k \sum_{s=1}^{k-1} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k-s}}^k \frac{F(s)}{k} \frac{F(k_1) \dots F(k_n)}{(k-s)(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{F(k)}{k} \\
&= \sum_{n=1}^{k-1} \sum_{k_1=1}^{k-1} \sum_{\substack{k_2, \dots, k_{n+1}=1 \\ \text{s.t. } \sum_{j=2}^{n+1} k_j = k-k_1}}^k \frac{F(k_1)}{k} \frac{F(k_2) \dots F(k_{n+1})}{\underbrace{(k-k_1)(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_{h+1})}_{\sum_{h=2}^{n+1} k_h}} + \frac{F(k)}{k} \\
&= \sum_{n=1}^{k-1} \sum_{k_1=1}^{k-1} \sum_{\substack{k_2, \dots, k_{n+1}=1 \\ \text{s.t. } \sum_{j=1}^{n+1} k_j = k}}^k \frac{F(k_1) F(k_2) \dots F(k_{n+1})}{k(\prod_{j=1}^n \sum_{h=n-j+2}^{n+1} k_h)} + \frac{F(k)}{k} \\
&= \sum_{n=2}^k \sum_{k_1=1}^{k-1} \sum_{\substack{k_2, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k}}^k \frac{F(k_1) F(k_2) \dots F(k_n)}{k(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{F(k)}{k} \\
&= \sum_{n=1}^k \sum_{k_1=1}^k \sum_{\substack{k_2, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j = k}}^k \frac{F(k_1) F(k_2) \dots F(k_n)}{k(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} = E_{(k)},
\end{aligned}$$

thus concluding the proof. $\square$

Note that the technical reason why we choose to assume F_t to be polynomial ($N < \infty$) instead of analytic is because in this proof we used in a couple of instances the property that $\int_0^t \sum_{k=0}^N s^k F_{(k+1)} ds = \sum_{k=0}^N \int_0^t s^k F_{(k+1)} ds$ which trivially holds if $N < \infty$. To lift this assumption we would need to assume some extra properties about the operators $F_{(k+1)}$ so that the order between the summation and integral can be changed.

Appendix B: Calculations for the spin boson model

We can write

$$H = \underbrace{\mathbb{1}_S}_{\equiv S_0} \otimes \underbrace{H_B}_{\equiv E_0} + \underbrace{\sigma_z}_{S_1} \otimes \underbrace{\left(\frac{g}{2} \mathbb{1} + \sum_k \lambda_k b_k + \bar{\lambda}_k b_k^\dagger \right)}_{\equiv E_1}$$

so that the second order term becomes $F_{(2)}(\mu) = \sum_{j,k=0}^1 \chi_{j,k}(\rho_B) \left(S_j \mu S_k - \frac{1}{2} \{S_j^\dagger S_k, \mu\} \right)$ where $\chi_{j,k}(\rho_B) = \langle E_k E_j \rangle_B - \langle E_j \rangle_B \langle E_k \rangle_B$. We can then start by noting that $\langle E_1 \rangle_B = g/2$ and

$$\langle H_B \rangle_B = \sum_{k=1}^{N_B} \omega_k \left(\langle b_k^\dagger b_k \rangle_B + \langle \mathbb{1} \rangle_B / 2 \right) = \sum_{k=1}^{N_B} \omega_k \left(\frac{1}{e^{\beta \omega_k} - 1} + \frac{1}{2} \right) = \sum_{k=1}^{N_B} \frac{\omega_k}{2} \frac{e^{\beta \omega_k} + 1}{e^{\beta \omega_k} - 1}$$

in order to compute

$$\begin{aligned}
\chi_{1,0}(\rho_B) &= \left\langle \left(\frac{g}{2} \mathbb{1} + \sum_j \lambda_j b_j + \bar{\lambda}_j b_j^\dagger \right) \sum_k \omega_k \left(n_k + \frac{1}{2} \right) \right\rangle_B - \frac{g}{2} \langle H_B \rangle_B \\
&= \sum_k \frac{g}{2} \omega_k (\langle n_k \rangle_B + \frac{1}{2}) + \sum_j \lambda_j \omega_k (\langle b_j n_k \rangle_B + \langle b_j \rangle_B) + \bar{\lambda}_j \omega_k (\langle b_j^\dagger n_k \rangle_B + \langle b_j^\dagger \rangle_B) - \frac{g}{2} \langle H_B \rangle_B \\
&= \sum_k \frac{g}{2} \omega_k \left(\frac{1}{e^{\beta \omega_k} - 1} + \frac{1}{2} \right) - \frac{g}{2} \frac{\omega_k}{2} \frac{e^{\beta \omega_k} + 1}{e^{\beta \omega_k} - 1} = 0
\end{aligned}$$

as $\langle b_j n_k \rangle_B = \langle b_j^\dagger n_k \rangle_B = 0$.

This fact directly implies that $F_{(2)}(\mu) = \sum_{j=0}^1 \chi_{j,j}(\rho_B) \mathcal{D}_{S_j}(\mu)$ and, since $S_0 = \mathbb{1}_S$ we have that $\mathcal{D}_{S_0} = 0$ thus recovering $F_{(2)}(\mu) = \chi_{1,1}(\rho_B) \mathcal{D}_{\sigma_z}(\mu)$. It then remains to compute the coefficient

$$\begin{aligned}
\chi_{1,1}(\rho_B) &= \left\langle \left(\frac{g}{2} \mathbb{1} + \sum_k \lambda_k b_k + \bar{\lambda}_k b_k^\dagger \right) \left(\frac{g}{2} \mathbb{1} + \sum_k \lambda_k b_k + \bar{\lambda}_k b_k^\dagger \right) \right\rangle_B - \frac{g^2}{4} \\
&= \cancel{\frac{g^2}{4} \langle \mathbb{1} \rangle_B} + \sum_k g \left(\cancel{\lambda_k \langle b_k \rangle_B} + \bar{\lambda}_k \cancel{\langle b_k^\dagger \rangle_B} \right) \\
&\quad + \sum_{j,k} \left(\lambda_j \lambda_k \cancel{\langle b_j b_k \rangle_B} + \lambda_j \bar{\lambda}_k \langle b_j b_k^\dagger \rangle_B + \bar{\lambda}_j \lambda_k \langle b_j^\dagger b_k \rangle_B + \bar{\lambda}_j \bar{\lambda}_k \cancel{\langle b_j^\dagger b_k^\dagger \rangle_B} \right) - \cancel{\frac{g^2}{4}} \\
&= \sum_{k=1}^{N_B} \frac{2|\lambda_k|^2}{e^{\beta\omega_k} - 1} \equiv \varphi.
\end{aligned}$$